

Same- i Leakage: A Resource Theory of Shared Complex Structure in Real Quantum Composition

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Complex quantum theory can be represented over real Hilbert spaces by replacing multiplication by i with an orthogonal complex structure J satisfying $J^2 = -I$ and $J^\top = -J$. For a single system this makes the imaginary unit appear locally removable. We show that the nontrivial structure reappears under composition: the full real tensor product of two realified systems is twice as large as the realification of the ordinary complex tensor product, and ordinary complex composition is recovered by imposing the same- i balancing relation $J_A x \otimes y = x \otimes J_B y$. Equivalently, the real tensor product splits into an aligned sector $H_+ = \ker(J_A \otimes J_B + I)$, representing ordinary complex composition, and an anti-aligned leakage sector $H_- = \ker(J_A \otimes J_B - I)$.

We define same- i leakage $L(\rho) = \text{Tr}(P_- \rho)$ and develop it into a resource theory. The headline results are the coherent cross-sector bound $\|P_+ \rho P_-\|_1 \leq \sqrt{L(1-L)}$, saturated by pure coherent cross-sector states, and an exactly solved sector-parity channel discrimination task with optimum $p_{\text{succ}}^{(\epsilon)} = \frac{1}{2} + \sqrt{m(1-m)}$, $m = \min\{\epsilon, 1/2\}$. We characterize the leakage-nonincreasing operations by the exact condition $\Phi^\dagger(P_-) \leq P_-$, prove a Kraus no-leakage criterion $P_- K_k P_+ = 0$, derive Hamiltonian and Lindblad leakage-rate formulas, and show that the trace distance to the free face equals $\sqrt{L}$ on coherent pure states while standard positive-noise robustness diverges for every state with $L > 0$.

As a worked example, the Caves–Fuchs–Rungta two-rebit state $\frac{1}{4}(\sigma_0 \otimes \sigma_0 + \alpha \sigma_y \otimes \sigma_y)$ has leakage $L = (1-\alpha)/2$, so the maximally real-entangled, complex-separable state sits at the center of the free face. We identify the explicit obstruction $L_B = 1/2$ for naive Bob-node leakage witnesses in product-source bilocal models. The framework gives a resource theory of compositional complex-structure alignment and isolates the construction of a non-tautological network witness, under operational rather than product-state source independence, as the central open problem.

1 Introduction

1.1 Why composition is where complex structure becomes nontrivial

A finite-dimensional complex Hilbert space $H_{\mathbb{C}}$ of complex dimension d can be reinterpreted as a real Hilbert space $H_{\mathbb{R}}$ of real dimension $2d$, together with a real-linear operator J acting as multiplication by i :

$$J^2 = -I, \quad J^\top = -J. \quad (1)$$

This is standard; the construction goes back to Stueckelberg [14] and is developed in detail by Myrheim [11] and Moretti–Oppio [10] at the level of single systems.

The substantive question is not whether a single complex system admits a real reformulation—it does—but whether two *independently* realified systems share the same local meaning of i . The ordinary complex tensor product imposes the balancing relation

$$J_A x \otimes y = x \otimes J_B y, \quad (2)$$

which says that multiplying subsystem A by i is the same as multiplying subsystem B by i . Without that constraint, the full real tensor product is twice as large as the realification of the complex tensor product, and the extra degrees of freedom describe configurations in which the two subsystems disagree about the complex structure. This paper quantifies that disagreement and develops it into a resource theory.

1.2 What this paper proves

We define the *alignment operator* $M = J_A \otimes J_B$. It is a real symmetric involution ($M^\top = M$, $M^2 = I$), and induces the decomposition

$$K_{AB} = H_{\mathbb{R}}^A \otimes_{\mathbb{R}} H_{\mathbb{R}}^B = H_+ \oplus H_-, \quad P_+ = \frac{1}{2}(I - M), \quad P_- = \frac{1}{2}(I + M). \quad (3)$$

The labeling places the aligned “physical” sector at M -eigenvalue -1 : $H_+ = \ker(M + I)$ is the subspace on which $\mathbf{A}\psi = \mathbf{B}\psi$ (Theorem 3), and $P_+ = \frac{1}{2}(I - M)$ projects onto it. The aligned sector H_+ canonically represents the same- i quotient and hence ordinary complex composition (Theorems 2 and 3). We define

$$L(\rho) := \text{Tr}(P_- \rho) = \frac{1}{2}(1 + \text{Tr}(M\rho)), \quad (4)$$

and prove:

- L is affine, ranges in $[0, 1]$, and vanishes exactly on states supported in H_+ (Theorem 5).
- A real CPTP map Φ is leakage-nonincreasing for all states iff $\Phi^\dagger(P_-) \leq P_-$ (Theorem 7).
- Kraus operators satisfy this iff $P_- K_k P_+ = 0$ for every k (Theorem 8).
- Coherent cross-sector amplitudes obey $\|P_+ \rho P_-\|_1 \leq \sqrt{L(1-L)}$ (Theorem 6).
- The trace distance to the free face equals $\sqrt{\lambda}$ on coherent pure states (Theorem 9).
- Standard positive-noise robustness diverges for every state with $L > 0$ (Proposition 11).
- Hamiltonian and Lindblad dynamics admit explicit leakage-rate formulas (Theorems 12, 13, and 14).
- Binary discrimination of $\mathcal{N}_0(\rho) = \rho$ versus $\mathcal{N}_1(\rho) = M\rho M$ has optimum $\frac{1}{2} + \sqrt{m(1-m)}$ with $m = \min\{\epsilon, 1/2\}$, achieved by coherent cross-sector pure states (Theorem 16).
- The Caves–Fuchs–Rungta two-rebit family has $L = (1 - \alpha)/2$ (Proposition 19).
- For product-source bilocal networks, Bob-node leakage is identically $1/2$ (Proposition 20).
- The leakage-constrained moment hierarchy is sound (Theorem 21).

1.3 What this paper does *not* claim

We do not claim that complex numbers are experimentally necessary, nor that real Hilbert-space quantum theory is falsified. Renou et al. [12] argued the former under product-state source independence; Hoffreumon–Woods [8, 7] and Weilenmann–Gisin–Sekatski [16] argue the opposite under operational or partial-independence variants. Our contribution is orthogonal: we quantify, as a resource, the compositional cost of enforcing one shared complex structure across independently composed real systems. We do not provide a non-tautological device-independent network inequality, and we explicitly prove an obstruction (Proposition 20) showing why the natural Bob-node leakage witness fails for product bilocal sources. We also do not claim that same- i leakage is operationally separated from ordinary Hickey–Gour imaginarity [6, 17, 18] in full generality; the inequivalence is structural and example-based (Section 8).

1.4 Positioning

The sector decomposition $K_{AB} = H_+ \oplus H_-$ has appeared implicitly in the realification literature beginning with Myrheim [11] and is anchored on the real-vs-complex composition phenomenon of Caves–Fuchs–Rungta [3]; our contribution is the resource-theoretic structure on top of it—a monotone, free operations, a coherent leakage bound, dynamics, an exact discrimination task, and an explicit obstruction for naive Bob-node bilocal witnesses. The framing is distinct from the Hickey–Gour basis-relative imaginarity resource theory [6, 17, 18], from the ubit construction of Aleksandrova–Borish–Wootters [1], from the real-vs-complex entanglement gap of Chiribella et al. [4], and from the Renou / Hoffreumon–Woods / Weilenmann–Gisin–Sekatski falsifiability axis. It also differs from symmetry-covariant resource theories of asymmetry in the sense of Marvian–Spekkens: the alignment operator M does not generate a group action whose orbit defines the free set, but is an order-two involution whose eigenspaces directly partition the state space. Resource-theoretically this is a sector-grading construction (the free face is a proper face of the state space, not a symmetry-orbit-stabilizing set), which is exactly what forces the standard positive-noise robustness to diverge (Proposition 11). Among the bullets in Section 1.5 below, each result lists the section that develops it, so the reader can navigate directly.

1.5 Roadmap

Section 2 sets up the realification map R and proves Theorem 1. Section 3 establishes the same- i quotient and sector decomposition (Theorems 2, 3). Section 4 defines L and proves the four core resource-theoretic theorems (5–8). Section 5 develops the geometry (Theorem 9 and Propositions 10–11). Section 6 treats dynamics (Theorems 12–14). Section 7 solves the sector-parity discrimination task (Theorem 16). Section 8 shows formal inequivalence with Hickey–Gour imaginarity via two explicit examples. Section 9 computes the Caves–Fuchs–Rungta family (Proposition 19). Section 10 proves the Bob-node bilocal obstruction (Proposition 20). Section 11 states moment-matrix soundness (Theorem 21). Section 12 discusses limitations and open problems.

1.6 Notation summary

All Hilbert spaces are finite-dimensional. The real adjoint is $\top$; in an orthonormal real basis this is transpose. Complex Hilbert spaces are $H_{\mathbb{C}}$ of complex dimension d , with realifications $H_{\mathbb{R}}$ of real dimension $2d$. The local complex structures are J_A, J_B (each satisfies $J^2 = -I$, $J^\top = -J$). The *sector operators* are

$$A := J_A \otimes I_B, \quad B := I_A \otimes J_B, \quad M := AB = J_A \otimes J_B. \quad (5)$$

The Schrödinger real generator is $G = -JH_{\mathbb{R}}$ where $H_{\mathbb{R}}$ is the realified Hamiltonian. Lindblad jump operators are R_ℓ . The trace norm is $\|\cdot\|_1$. Density operators ρ are real symmetric positive semidefinite with $\text{Tr } \rho = 1$. The free face is

$$\mathcal{F}_{AB} := \{\rho : L(\rho) = 0\} = \{\rho : \rho = P_+ \rho P_+\}. \quad (6)$$

2 Realification and local complex structures

2.1 Orthogonal complex structures

Theorem 1 (Realification preserves quantum statistics). *Let $X = A + iB$ be a $d \times d$ complex matrix with real A, B , and define*

$$R(X) := \begin{pmatrix} A & -B \\ B & A \end{pmatrix}. \quad (7)$$

Let $\rho_{\mathbb{C}}$ be a complex density operator and $E_{\mathbb{C}}$ a complex effect. Define $\rho_{\mathbb{R}} := \frac{1}{2}R(\rho_{\mathbb{C}})$ and $E_{\mathbb{R}} := R(E_{\mathbb{C}})$. Then

$$\text{Tr}_{\mathbb{R}} \rho_{\mathbb{R}} = 1, \quad \text{Tr}_{\mathbb{R}}(E_{\mathbb{R}} \rho_{\mathbb{R}}) = \text{Tr}_{\mathbb{C}}(E_{\mathbb{C}} \rho_{\mathbb{C}}). \quad (8)$$

Proof. The map R is a unital ring homomorphism: $R(XY) = R(X)R(Y)$ by direct block computation. For $X = A + iB$, $\text{Tr}_{\mathbb{R}} R(X) = 2 \text{Tr} A = 2\Re \text{Tr}_{\mathbb{C}} X$. Since $\rho_{\mathbb{C}}$ is Hermitian, $\text{Tr} \rho_{\mathbb{C}} \in \mathbb{R}$ and equals 1, so $\text{Tr}_{\mathbb{R}} R(\rho_{\mathbb{C}}) = 2$, giving $\text{Tr}_{\mathbb{R}} \rho_{\mathbb{R}} = 1$. Therefore

$$\text{Tr}_{\mathbb{R}}(E_{\mathbb{R}}\rho_{\mathbb{R}}) = \frac{1}{2} \text{Tr}_{\mathbb{R}} R(E_{\mathbb{C}}\rho_{\mathbb{C}}) = \Re \text{Tr}_{\mathbb{C}}(E_{\mathbb{C}}\rho_{\mathbb{C}}) = \text{Tr}_{\mathbb{C}}(E_{\mathbb{C}}\rho_{\mathbb{C}}),$$

where the last equality uses that the trace of a product of Hermitian operators is real. $\square$

The *canonical* complex structure on $\mathbb{R}^{2d}$ is

$$J_d := \begin{pmatrix} 0 & -I_d \\ I_d & 0 \end{pmatrix}, \quad (9)$$

which satisfies $J_d^2 = -I$ and $J_d^{\top} = -J_d$. The induced complex inner product is $\langle \psi, \phi \rangle_J = g(\psi, \phi) + i g(J\psi, \phi)$, where g is the real inner product. A real-linear operator X on $(H_{\mathbb{R}}, J)$ is complex-linear iff $XJ = JX$.

3 same- i composition

3.1 The quotient theorem

Theorem 2 (same- i quotient). *Let $H_{\mathbb{C}}^A, H_{\mathbb{C}}^B$ be complex Hilbert spaces of complex dimensions d_A, d_B , with realifications $H_{\mathbb{R}}^A, H_{\mathbb{R}}^B$ carrying complex structures J_A, J_B . There is a canonical real-linear isomorphism*

$$(H_{\mathbb{C}}^A \otimes_{\mathbb{C}} H_{\mathbb{C}}^B)_{\mathbb{R}} \cong \frac{H_{\mathbb{R}}^A \otimes_{\mathbb{R}} H_{\mathbb{R}}^B}{\text{span}_{\mathbb{R}}\{J_A x \otimes y - x \otimes J_B y : x \in H_{\mathbb{R}}^A, y \in H_{\mathbb{R}}^B\}}. \quad (10)$$

Proof. The complex tensor product is the universal target of complex-bilinear maps. A real-bilinear map $\varphi : H_{\mathbb{R}}^A \times H_{\mathbb{R}}^B \rightarrow V$ into a complex vector space V factors through the complex tensor product iff $\varphi(J_A x, y) = \varphi(x, J_B y)$ for all x, y . Equivalently, φ vanishes on the balancing relations $J_A x \otimes y - x \otimes J_B y$. The quotient on the right-hand side therefore satisfies the universal property of $(H_{\mathbb{C}}^A \otimes_{\mathbb{C}} H_{\mathbb{C}}^B)_{\mathbb{R}}$. $\square$

3.2 Sector decomposition

Define the sector operators $A = J_A \otimes I_B$, $B = I_A \otimes J_B$, and the alignment operator $M = AB = J_A \otimes J_B$. Then M is a real symmetric involution: $M^{\top} = J_A^{\top} \otimes J_B^{\top} = (-J_A) \otimes (-J_B) = M$, and $M^2 = J_A^2 \otimes J_B^2 = (-I)(-I) = I$. The real tensor product $K_{AB} = H_{\mathbb{R}}^A \otimes_{\mathbb{R}} H_{\mathbb{R}}^B$ has $\dim_{\mathbb{R}} K_{AB} = 4d_A d_B$, exactly twice the real dimension of the complex tensor product. Define $H_+ = \ker(M + I)$, $H_- = \ker(M - I)$, and $P_+ = \frac{1}{2}(I - M)$, $P_- = \frac{1}{2}(I + M)$.

Theorem 3 (Sector representation of the quotient). *The aligned sector H_+ canonically represents the same- i quotient; equivalently, $\psi \in H_+$ iff $A\psi = B\psi$. The sector has real dimension $\dim_{\mathbb{R}} H_+ = 2d_A d_B$ and carries the induced complex structure*

$$J_{AB} := A|_{H_+} = B|_{H_+}, \quad J_{AB}^2 = -I_{H_+}. \quad (11)$$

Proof. Suppose $A\psi = B\psi$. Applying A gives $A^2\psi = AB\psi = M\psi$. Since $A^2 = J_A^2 \otimes I = -I$, we have $M\psi = -\psi$, hence $\psi \in H_+$. Conversely, $M\psi = -\psi$ gives $AB\psi = -\psi$, so $B\psi = -A^{-1}\psi = A\psi$ (using $A^{-1} = -A$). On H_+ , $J_{AB}^2 = A^2 = -I$. The dimension count $2d_A d_B$ follows from $\dim H_+ + \dim H_- = 4d_A d_B$ together with $\text{Tr}(M) = \text{Tr}(J_A) \text{Tr}(J_B) = 0$ (each J is antisymmetric), which forces $\dim H_+ = \dim H_-$. $\square$

3.3 Recovery of ordinary complex quantum theory

A real density operator ρ on K_{AB} corresponds to a state of ordinary complex quantum theory on $H_{\mathbb{C}}^A \otimes_{\mathbb{C}} H_{\mathbb{C}}^B$ iff

$$\rho = P_+ \rho P_+ \quad \text{and} \quad [\rho, J_{AB}] = 0, \quad (12)$$

and similarly for effects. The first condition is zero leakage, $L(\rho) = 0$; the second is J_{AB} -compatibility on the aligned sector. These are independent constraints. This paper develops the resource theory of the *first* condition.

4 same- i leakage as a resource

4.1 Definition and basic properties

Definition 4. For a real density operator ρ on K_{AB} ,

$$L(\rho) := \text{Tr}(P_- \rho) = \frac{1}{2}(1 + \text{Tr}(M\rho)). \quad (13)$$

The *free face* is $\mathcal{F}_{AB} = \{\rho : L(\rho) = 0\}$.

Theorem 5 (Basic leakage properties). *For every real density operator ρ ,*

$$0 \leq L(\rho) \leq 1, \quad L(\rho) = 0 \iff \rho = P_+ \rho P_+, \quad (14)$$

and L is affine: $L(\sum_j p_j \rho_j) = \sum_j p_j L(\rho_j)$.

Proof. Since $0 \leq P_- \leq I$, $0 \leq \text{Tr}(P_- \rho) \leq \text{Tr} \rho = 1$. If $L(\rho) = 0$, then $\text{Tr}(P_- \rho P_-) = 0$ and $P_- \rho P_- \succeq 0$ forces $P_- \rho P_- = 0$. Writing ρ in block form on $H_+ \oplus H_-$,

$$\rho = \begin{pmatrix} \rho_{++} & \rho_{+-} \\ \rho_{-+} & \rho_{--} \end{pmatrix},$$

$\rho_{--} = 0$ together with $\rho \succeq 0$ forces $\rho_{+-} = 0$ by the Schur complement / Douglas factorization. Affineness is immediate from trace linearity. $\square$

4.2 Coherent leakage bound

Theorem 6 (Coherent leakage bound). *For every real density operator ρ ,*

$$\|P_+ \rho P_-\|_1 \leq \sqrt{L(\rho)(1 - L(\rho))}. \quad (15)$$

The bound is saturated by pure coherent cross-sector states.

Proof. By the Douglas factorization for PSD block matrices [2], there exists a contraction K with $\|K\|_\infty \leq 1$ such that

$$\rho_{+-} = \rho_{++}^{1/2} K \rho_{--}^{1/2}.$$

By the multiplicative trace-norm bound,

$$\|\rho_{+-}\|_1 \leq \|\rho_{++}^{1/2}\|_2 \|K\|_\infty \|\rho_{--}^{1/2}\|_2 \leq \sqrt{\text{Tr} \rho_{++}} \sqrt{\text{Tr} \rho_{--}} = \sqrt{(1 - L)L}.$$

For the pure state $|\psi\rangle = \sqrt{1 - \lambda}|\psi_+\rangle + \sqrt{\lambda}|\psi_-\rangle$ with $|\psi_\pm\rangle \in H_\pm$ unit, direct computation gives $L = \lambda$ and $\|P_+|\psi\rangle\langle\psi|P_-\|_1 = \sqrt{\lambda(1 - \lambda)}$. $\square$

4.3 Free operations

Theorem 7 (Leakage monotonicity iff). *A real CPTP map Φ on K_{AB} satisfies $L(\Phi(\rho)) \leq L(\rho)$ for every density operator ρ iff $\Phi^\dagger(P_-) \leq P_-$.*

Proof. $(\Leftarrow)$ $L(\Phi(\rho)) = \text{Tr}(P_- \Phi(\rho)) = \text{Tr}(\Phi^\dagger(P_-) \rho) \leq \text{Tr}(P_- \rho) = L(\rho)$. $(\Rightarrow)$ Since Φ is real CP and P_- is real symmetric, $\Phi^\dagger(P_-)$ is real symmetric. The inequality $\text{Tr}((\Phi^\dagger(P_-) - P_-)\rho) \leq 0$ for every real density ρ forces $\Phi^\dagger(P_-) - P_- \leq 0$ as a real symmetric operator (test on rank-one $\rho = |v\rangle\langle v|$ for arbitrary real unit v , giving $\langle v | (\Phi^\dagger(P_-) - P_-) | v \rangle \leq 0$). $\square$

Theorem 8 (Kraus no-leakage characterization). *For a real CP map $\Phi(\rho) = \sum_k K_k \rho K_k^\top$ satisfying the trace-preservation identity $\sum_k K_k^\top K_k = I$, the dual condition $\Phi^\dagger(P_-) \leq P_-$ holds iff*

$$P_- K_k P_+ = 0 \quad \text{for every } k. \quad (16)$$

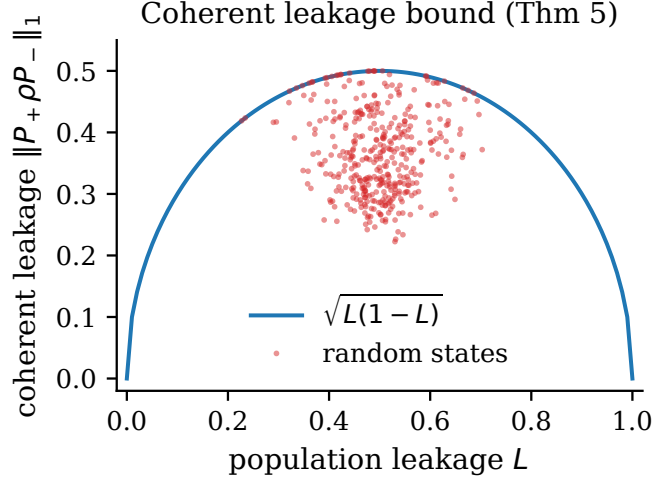


Figure 1: Coherent leakage $\|P_+ \rho P_-\|_1$ against population leakage $L(\rho)$. Dots: 400 random real density operators of varying rank on K_{AB} with $d_A = d_B = 2$. Curve: the saturation envelope $\sqrt{L(1-L)}$ from Theorem 6. The bound is saturated only by pure coherent cross-sector states.

Proof. ($\Rightarrow$) Compressing $\Phi^\dagger(P_-) \leq P_-$ to H_+ :

$$P_+ \Phi^\dagger(P_-) P_+ = \sum_k (P_- K_k P_+)^T (P_- K_k P_+) \leq P_+ P_- P_+ = 0.$$

Each summand is PSD and sums to zero, so each vanishes.

($\Leftarrow$) In block form

$$K_k = \begin{pmatrix} K_{++ ,k} & K_{+- ,k} \\ 0 & K_{-- ,k} \end{pmatrix}, \quad K_k^T P_- K_k = \begin{pmatrix} 0 & 0 \\ 0 & K_{-- ,k}^T K_{-- ,k} \end{pmatrix}.$$

Trace preservation gives $\sum_k (K_{+- ,k}^T K_{+- ,k} + K_{-- ,k}^T K_{-- ,k}) = I_-$, hence $\sum_k K_{-- ,k}^T K_{-- ,k} \leq I_-$ and $\Phi^\dagger(P_-) \leq P_-$. $\square$

A channel is *strongly sector-preserving* if additionally $[K_k, M] = 0$ for every k . The block decomposition then forces $K_{+- ,k} = 0$ and yields exact conservation $L(\Phi(\rho)) = L(\rho)$.

Remark (Reduction at $d_A = 1$). The construction does *not* trivialize at $d_A = 1$. With $d_A = 1$ and d_B arbitrary, $H_{\mathbb{R}}^A = \mathbb{R}^2$ with J_A a 2×2 real antisymmetric square root of $-I$, and $K_{AB} = \mathbb{R}^2 \otimes_{\mathbb{R}} \mathbb{R}^{2d_B}$ of real dimension $4d_B$. The alignment operator $M = J_A \otimes J_B$ has $\text{Tr } M = \text{Tr}(J_A) \cdot \text{Tr}(J_B) = 0$ (each J is antisymmetric), so $\dim H_+ = \dim H_- = 2d_B$ exactly. In particular the Caves–Fuchs–Rungta family (Section 9) sits at $d_A = d_B = 1$, where $\dim H_+ = \dim H_- = 2$ and the entire leakage spectrum $[0, 1]$ is realized by varying the parameter α .

5 Geometry of the resource

5.1 Trace distance to the free face

Theorem 9 (Trace distance on coherent pure states). *Let $|\psi\rangle = \sqrt{1-\lambda}|\psi_+\rangle + \sqrt{\lambda}|\psi_-\rangle$ with $|\psi_{\pm}\rangle \in H_{\pm}$ unit and $0 \leq \lambda \leq 1$. Then*

$$D_1(|\psi\rangle\langle\psi|, \mathcal{F}_{AB}) := \inf_{\sigma \in \mathcal{F}_{AB}} \frac{1}{2} \|\ |\psi\rangle\langle\psi| - \sigma \|_1 = \sqrt{\lambda}. \quad (17)$$

Proof. Upper bound. Take $\sigma = |\psi_+\rangle\langle\psi_+| \in \mathcal{F}_{AB}$. The trace distance between two pure states equals $\sqrt{1 - |\langle\psi|\psi_+\rangle|^2} = \sqrt{1 - (1-\lambda)} = \sqrt{\lambda}$.

Lower bound. Let $\sigma \in \mathcal{F}_{AB}$. Decompose σ along the rank-one projector $|\psi_+\rangle\langle\psi_+|$ and its orthogonal complement *inside* H_+ : write $\sigma = \alpha|\psi_+\rangle\langle\psi_+| + (1-\alpha)\tau$ with $\alpha \in [0, 1]$ and τ a density operator supported in $H_+ \ominus \text{span}\{\psi_+\}$. The supports of $|\psi\rangle\langle\psi| - \alpha|\psi_+\rangle\langle\psi_+|$ (lives in $\text{span}\{\psi_+, \psi_-\}$) and $(1-\alpha)\tau$ (orthogonal to that span) are mutually orthogonal, so

$$\| |\psi\rangle\langle\psi| - \sigma \|_1 = \| |\psi\rangle\langle\psi| - \alpha|\psi_+\rangle\langle\psi_+| \|_1 + (1-\alpha).$$

In the basis $\{\psi_+, \psi_-\}$,

$$|\psi\rangle\langle\psi| - \alpha|\psi_+\rangle\langle\psi_+| = \begin{pmatrix} 1-\lambda-\alpha & \sqrt{\lambda(1-\lambda)} \\ \sqrt{\lambda(1-\lambda)} & \lambda \end{pmatrix},$$

with trace $1-\alpha$ and determinant $-\alpha\lambda$. Its eigenvalues are $\frac{1-\alpha}{2} \pm \frac{1}{2}\sqrt{(1-\alpha)^2 + 4\alpha\lambda}$, so its trace norm equals $\sqrt{(1-\alpha)^2 + 4\alpha\lambda}$. Therefore

$$\frac{1}{2}\| |\psi\rangle\langle\psi| - \sigma \|_1 = \frac{1}{2} \left[\sqrt{(1-\alpha)^2 + 4\alpha\lambda} + (1-\alpha) \right] =: f(\alpha).$$

A direct check ($f'(\alpha) = 0$ only at $\lambda \in \{0, 1\}$, and $f(0) = 1$ while $f(1) = \sqrt{\lambda}$) shows f is strictly decreasing on $[0, 1]$ for every $\lambda \in (0, 1)$, hence is minimized at $\alpha = 1$ with value $\sqrt{\lambda}$. Combined with the upper bound this gives equality. $\square$

5.2 Alignment weight

The convex-roof alignment weight is

$$W(\rho) := \inf\{\lambda \in [0, 1] : \rho = (1-\lambda)\sigma + \lambda\tau, \sigma \in \mathcal{F}_{AB}, \tau \in \mathcal{D}(K_{AB})\}. \quad (18)$$

Proposition 10 (Alignment weight ordering). *$W(\rho) \geq L(\rho)$, with equality on states block-diagonal in $H_+ \oplus H_-$. For coherent pure cross-sector states with $0 < \lambda < 1$, $W = 1$ while $L = \lambda$.*

Proof. If $\rho = (1-\lambda)\sigma + \lambda\tau$ with σ free, then $L(\rho) = \lambda L(\tau) \leq \lambda$; taking the infimum gives $L \leq W$. For block-diagonal ρ , the decomposition $\rho = (\rho_{++} \oplus 0) + (0 \oplus \rho_{--})$ realizes $W = L$. For a coherent pure state, extremality forces the decomposition to be trivial ($\sigma = |\psi\rangle\langle\psi|$ is not free), so $W = 1$. $\square$

5.3 Robustness diverges

Proposition 11 (Divergent positive-noise robustness). *The standard generalized robustness*

$$R(\rho) := \inf\{s \geq 0 : (\rho + s\tau)/(1+s) \in \mathcal{F}_{AB}, \tau \in \mathcal{D}(K_{AB})\} \quad (19)$$

is infinite whenever $L(\rho) > 0$.

Proof. If $(\rho + s\tau)/(1+s) \in \mathcal{F}_{AB}$, then $P_-(\rho + s\tau)P_- = 0$. Both $P_-\rho P_-$ and $P_-\tau P_-$ are PSD, and their sum vanishes only if each vanishes. Hence $L(\rho) = 0$. $\square$

The relevant operational quantifier in this resource theory is therefore the trace-distance witness D_1 , not the generalized robustness. This places the theory outside the standard robustness-quantifies-advantage framework of Takagi–Regula [15] and motivates the trace-distance-based geometric analysis above, in line with general resource-theoretic constructions where δ_1 -type measures replace robustness on non-convex-cone resources.

5.4 Object summary

6 Dynamics

6.1 Hamiltonian leakage

Let $H_{\mathbb{R}}$ be a real symmetric Hamiltonian on K_{AB} commuting with the global complex structure of the realified complex theory. The Schrödinger generator is $G = -JH_{\mathbb{R}}$, which is skew-symmetric ($G^\top = -G$), so the evolution $O(t) = e^{Gt}$ is orthogonal.

Theorem 12 (Hamiltonian leakage conservation). *If $[G, M] = 0$, then $L(\rho(t)) = L(\rho(0))$ for all t .*

Proof. $[G, M] = 0$ implies $[e^{Gt}, P_-] = 0$, so $\text{Tr}(P_- e^{Gt} \rho e^{-Gt}) = \text{Tr}(P_- \rho)$. $\square$

Object	Formula	Meaning
J_A, J_B	$J^2 = -I, J^\top = -J$	Local complex structures
M	$J_A \otimes J_B$	Alignment operator
$H_\pm$	$\ker(M \pm I)$	Aligned / anti-aligned sectors
$P_\pm$	$\frac{1}{2}(I \mp M)$	Sector projectors
$L(\rho)$	$\text{Tr}(P_- \rho)$	Population leakage
$C(\rho)$	$\ P_+ \rho P_- \ _1$	Coherent leakage
$W(\rho)$	convex-roof alignment weight	Mixture resource
$D_1(\rho, \mathcal{F}_{AB})$	$\inf_{\sigma \in \mathcal{F}_{AB}} \frac{1}{2} \ \rho - \sigma\ _1$	Trace distance to free face
$R(\rho)$	generalized robustness	$= \infty$ for $L > 0$

6.2 Perturbative leakage generation

Theorem 13 (Perturbative leakage). *Let $G_\epsilon = G_0 + \epsilon B$ with $G_0^\top = -G_0$, $B^\top = -B$, and $[G_0, M] = 0$. For $\psi_0 \in H_+$ unit and finite $t > 0$,*

$$L(|e^{G_\epsilon t} \psi_0\rangle\langle e^{G_\epsilon t} \psi_0|) = \epsilon^2 \left\| \int_0^t P_- e^{G_0(t-s)} B e^{G_0 s} \psi_0 ds \right\|^2 + O(\epsilon^3). \quad (20)$$

Proof. Duhamel expansion: $e^{G_\epsilon t} = e^{G_0 t} + \epsilon \int_0^t e^{G_0(t-s)} B e^{G_0 s} ds + O(\epsilon^2)$. Since $[G_0, M] = 0$, $e^{G_0 t} \psi_0 \in H_+$, so $P_- e^{G_0 t} \psi_0 = 0$. Therefore $P_- \psi(t) = \epsilon \int_0^t P_- e^{G_0(t-s)} B e^{G_0 s} \psi_0 ds + O(\epsilon^2)$, and $L = \|P_- \psi(t)\|^2 = O(\epsilon^2)$. $\square$

6.3 Lindblad leakage rate

For a real Lindblad generator with skew-symmetric Hamiltonian part G and jump operators R_ℓ ,

$$\mathcal{L}(\rho_t) := G \rho_t + \rho_t G^\top + \sum_\ell (R_\ell \rho_t R_\ell^\top - \frac{1}{2} \{R_\ell^\top R_\ell, \rho_t\}). \quad (21)$$

The dual generator is

$$\mathcal{L}^\dagger(X) = [X, G] + \sum_\ell (R_\ell^\top X R_\ell - \frac{1}{2} \{R_\ell^\top R_\ell, X\}). \quad (22)$$

Theorem 14 (Lindblad leakage rate). $\frac{d}{dt} L(\rho_t) = \text{Tr}(\mathcal{L}^\dagger(P_-) \rho_t)$. *If $\mathcal{L}^\dagger(P_-) = 0$, leakage is conserved; if $\mathcal{L}^\dagger(P_-) \leq 0$, leakage is nonincreasing.*

Corollary 15 (Sufficient conservation condition). *If $[G, M] = 0$ and $[R_\ell, M] = 0$ for every ℓ , then $\mathcal{L}^\dagger(P_-) = 0$ and $L(\rho_t) = L(\rho_0)$ for all t .*

Proof. $[G, M] = 0$ implies $[P_-, G] = 0$, so the Hamiltonian-commutator term in $\mathcal{L}^\dagger(P_-)$ vanishes. For each dissipator, $[R_\ell, M] = 0$ gives $[R_\ell, P_-] = 0$ and $[R_\ell^\top, P_-] = 0$, hence $R_\ell^\top R_\ell$ commutes with P_- . Then $R_\ell^\top P_- R_\ell = R_\ell^\top R_\ell P_- = P_- R_\ell^\top R_\ell = \frac{1}{2} \{R_\ell^\top R_\ell, P_-\}$, so the dissipator term vanishes. $\square$

Figure 2 verifies the $O(\epsilon^2)$ perturbative scaling of Theorem 13 numerically.

7 Sector-parity discrimination

Define the channels

$$\mathcal{N}_0(\rho) = \rho, \quad \mathcal{N}_1(\rho) = M \rho M. \quad (23)$$

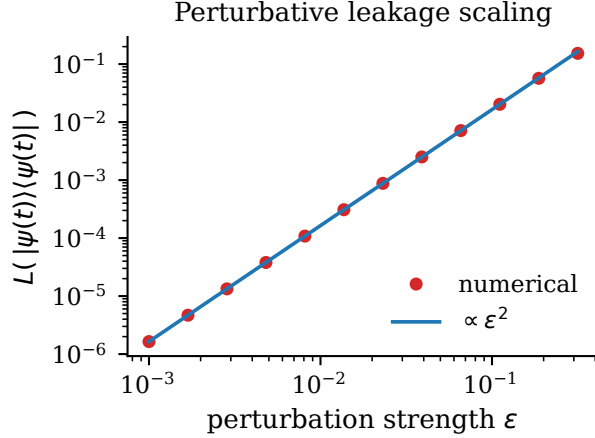


Figure 2: Leakage generated by a perturbation $G_0 + \epsilon B$ acting on an initially aligned ($L = 0$) state, with $[G_0, M] = 0$. The log-log slope is 2, confirming the Duhamel expansion of Theorem 13.

Theorem 16 (Sector-parity discrimination). *The optimal binary discrimination success probability with equal priors and leakage budget $L(\rho) \leq \epsilon$ is*

$$p_{\text{succ}}^{(\epsilon)} = \frac{1}{2} + \sqrt{m(1-m)}, \quad m = \min\{\epsilon, 1/2\}. \quad (24)$$

Equivalently, $p_{\text{succ}}^{(\epsilon)} = \frac{1}{2} + \sqrt{\epsilon(1-\epsilon)}$ for $0 \leq \epsilon \leq 1/2$, and $p_{\text{succ}}^{(\epsilon)} = 1$ for $\epsilon \geq 1/2$, saturated by coherent pure cross-sector states.

Proof. Using $M = P_- - P_+$,

$$\begin{aligned} M\rho M &= (P_- - P_+)\rho(P_- - P_+) \\ &= P_- \rho P_- - P_+ \rho P_- - P_- \rho P_+ + P_+ \rho P_+, \end{aligned}$$

so $\rho - M\rho M = 2(P_+ \rho P_- + P_- \rho P_+)$. The matrix $X := P_+ \rho P_- + P_- \rho P_+$ has the off-diagonal block form $\begin{pmatrix} 0 & B \\ B^\top & 0 \end{pmatrix}$ with $B := P_+ \rho P_-$. Its eigenvalues come in pairs $\pm s_i$ for s_i the singular values of B , so $\|X\|_1 = \sum_i |s_i| + \sum_i |-s_i| = 2 \sum_i s_i = 2\|B\|_1$. Hence $\|\rho - M\rho M\|_1 = 4\|P_+ \rho P_-\|_1$. By the Holevo–Helstrom theorem [5],

$$p_{\text{succ}}(\rho) = \frac{1}{2} + \frac{1}{4} \|(\mathcal{N}_0 - \mathcal{N}_1)(\rho)\|_1 = \frac{1}{2} + \|P_+ \rho P_-\|_1.$$

Maximizing over $\{\rho : L(\rho) \leq \epsilon\}$ using Theorem 6: the function $L \mapsto \sqrt{L(1-L)}$ is increasing on $[0, 1/2]$ and peaks at $L = 1/2$. For $\epsilon < 1/2$ the constraint is tight and the optimum is at $L = \epsilon$, giving $p_{\text{succ}}^{(\epsilon)} = \frac{1}{2} + \sqrt{\epsilon(1-\epsilon)}$; for $\epsilon \geq 1/2$ the unconstrained optimum $L = 1/2$ already satisfies the budget and gives $p_{\text{succ}}^{(\epsilon)} = 1$. The bound is saturated in both regimes by the pure coherent cross-sector state $|\psi\rangle = \sqrt{1-m}|\psi_+\rangle + \sqrt{m}|\psi_-\rangle$ with $m = \min\{\epsilon, 1/2\}$. $\square$

Block-diagonal probes give no advantage. For ρ block-diagonal on $H_+ \oplus H_-$, $P_+ \rho P_- = 0$ and $p_{\text{succ}}(\rho) = 1/2$. Population leakage alone provides no advantage; only coherent cross-sector amplitude does.

Limitation: imaginarity is not separated by this task. The optimal probe can be chosen with entrywise-real entries in the canonical basis where J_A, J_B are the standard symplectic matrices: take $\psi_+, \psi_- \in H_\pm$ unit and entrywise real (always possible because $H_\pm$ are real subspaces); then $|\psi\rangle\langle\psi|$ is entrywise real and has Hickey–Gour imaginarity zero in that basis, yet it achieves the maximal success probability. Therefore sector-parity discrimination witnesses coherent same- i leakage but does *not* separate it from ordinary basis imaginarity.

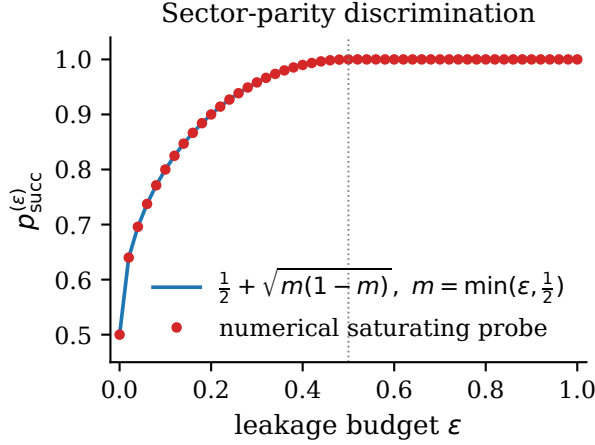


Figure 3: Sector-parity optimal success probability as a function of leakage budget ϵ , with the saturating coherent pure cross-sector probe constructed in the proof.

8 Inequivalence from imaginarity

The same- i leakage L and the Hickey–Gour basis-relative imaginarity $\mathcal{I}_B$ [6, 17] are structurally inequivalent. We give two explicit examples in the canonical basis.

Example 17 (Entrywise real, $L = 1/2$). Let $\psi_{\pm} \in H_{\pm}$ be entrywise-real unit vectors (possible because $H_{\pm}$ are real subspaces). Take $\psi = (\psi_+ + \psi_-)/\sqrt{2}$. Then $\rho = |\psi\rangle\langle\psi|$ has entirely real entries in the canonical basis—so its Hickey–Gour imaginarity is zero in that basis—but $L(\rho) = 1/2$.

Example 18 ($L = 0$, nonzero entrywise imaginarity). Let $|\psi\rangle_{\mathbb{C}} = (|00\rangle + i|11\rangle)/\sqrt{2}$ on $\mathbb{C}^2 \otimes_{\mathbb{C}} \mathbb{C}^2$. Its density operator $\rho_{\mathbb{C}} = |\psi\rangle_{\mathbb{C}}\langle\psi|_{\mathbb{C}}$ has $\rho_{\mathbb{C}}[0, 3] = -i/2$ and $\rho_{\mathbb{C}}[3, 0] = +i/2$ in the computational basis, so its entrywise imaginarity is nonzero. The canonical same- i embedding ι from Theorem 3—explicitly $\iota(|j\rangle_{\mathbb{C}} \otimes |k\rangle_{\mathbb{C}}) = (1/\sqrt{2})(r_j \otimes r_k - J_A r_j \otimes J_B r_k)$ on the canonical real basis—sends $|\psi\rangle_{\mathbb{C}}$ to a unit vector $\psi_R \in H_+ \subset K_{AB}$, and $\rho_R = |\psi_R\rangle\langle\psi_R|$ satisfies $L(\rho_R) = 0$ by construction.

These examples show formal inequivalence; an operational separation—a task where every L -resourceful state outperforms every $\mathcal{I}_B$ -free state, in every basis—remains open.

9 The Caves–Fuchs–Rungta example

Caves, Fuchs, and Rungta [3] introduced the two-rebit family

$$\rho_{\text{CFR}}(\alpha) = \frac{1}{4}(\sigma_0 \otimes \sigma_0 + \alpha \sigma_y \otimes \sigma_y), \quad -1 \leq \alpha \leq 1, \quad (25)$$

and showed it is real-entangled with real concurrence $|\alpha|$ but complex-separable. (Each σ_y has purely imaginary entries; their tensor product $\sigma_y \otimes \sigma_y$ has real entries, so $\rho_{\text{CFR}}(\alpha)$ is a bona fide real density matrix.) Embedding two rebits into the same- i framework with $J_A = J_B = i\sigma_y$, a 2×2 real antisymmetric square root of $-I$, we have

$$M = J_A \otimes J_B = (i\sigma_y) \otimes (i\sigma_y) = -\sigma_y \otimes \sigma_y, \quad (26)$$

so $\rho_{\text{CFR}}(\alpha) = \frac{1}{4}(I - \alpha M)$.

Proposition 19 (CFR leakage). $L(\rho_{\text{CFR}}(\alpha)) = (1 - \alpha)/2$.

Proof. $L = \text{Tr}(P_- \rho_{\text{CFR}}(\alpha)) = \frac{1}{8} \text{Tr}((I + M)(I - \alpha M)) = \frac{1}{8} \text{Tr}(I - \alpha M + M - \alpha M^2) = \frac{1}{8}(4 - 4\alpha) = (1 - \alpha)/2$, using $\text{Tr} M = 0$ (each $i\sigma_y$ is traceless and antisymmetric), $M^2 = I$, and $\text{Tr} I_4 = 4$. $\square$

Interpretation. At $\alpha = 1$, the maximally real-entangled CFR state (complex-separable) has $L = 0$: fully aligned, $\rho = \frac{1}{2}P_+$. At $\alpha = 0$, the maximally mixed state has $L = 1/2$. At $\alpha = -1$, leakage is $L = 1$: fully anti-aligned, $\rho = \frac{1}{2}P_-$. The CFR parameter interpolates linearly through the alignment axis (Figure 4). Key consequence: same- i leakage and real-vs-complex entanglement detect different things—the $\alpha = 1$ state is maximally real-entangled yet sits at the center of the free face.

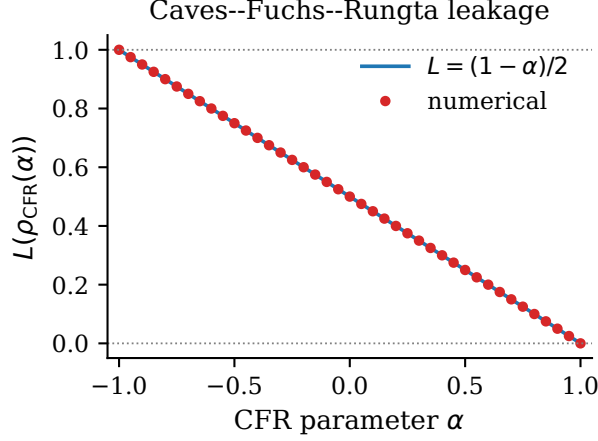


Figure 4: Leakage of the Caves–Fuchs–Rungta family $\rho_{\text{CFR}}(\alpha)$ as a function of α . The closed-form prediction $L = (1 - \alpha)/2$ from Proposition 19 is shown together with the direct numerical evaluation $L = \text{Tr}(P_- \rho_{\text{CFR}}(\alpha))$.

10 Bob-node obstruction in bilocal networks

Consider the bilocal scenario $A \leftarrow S_1 \rightarrow B \leftarrow S_2 \rightarrow C$, with independent product sources $\rho_{\text{net}} = \rho_{AB_1} \otimes_{\mathbb{R}} \rho_{B_2C}$ on the relevant real Hilbert spaces. The natural Bob-node alignment operator is $M_B = J_{B_1} \otimes J_{B_2}$, and one might hope to define a Bob-node leakage $L_B(\rho_{B_1B_2}) := \text{Tr}(P_-^B \rho_{B_1B_2})$ as a network witness, where $\rho_{B_1B_2} = \text{Tr}_{AC} \rho_{\text{net}}$ and $P_-^B = \frac{1}{2}(I + M_B)$.

Proposition 20 (Bob-node obstruction). *For every product bilocal source $\rho_{B_1B_2} = \rho_{B_1} \otimes_{\mathbb{R}} \rho_{B_2}$ with ρ_{B_i} real symmetric PSD,*

$$L_B(\rho_{B_1B_2}) = \frac{1}{2} \quad \text{exactly.} \quad (27)$$

Proof. $\text{Tr}(M_B \rho_{B_1} \otimes \rho_{B_2}) = \text{Tr}(J_{B_1} \rho_{B_1}) \cdot \text{Tr}(J_{B_2} \rho_{B_2})$. For any real symmetric ρ and real antisymmetric J , $\text{Tr}(J\rho) = \text{Tr}((J\rho)^\top) = \text{Tr}(\rho^\top J^\top) = -\text{Tr}(J\rho)$, so $\text{Tr}(J\rho) = 0$. Hence each factor vanishes and $L_B = \frac{1}{2}(1 + 0) = \frac{1}{2}$. $\square$

Consequence. The naive Bob-node leakage witness is structurally tautological for product-source bilocal models: it is fixed at $1/2$ regardless of the choice of source state, and therefore cannot detect any feature of the network. A non-tautological network witness requires at least one of the following: (i) replacing product-state independence with operational independence in the sense of Hoffreumon–Woods [8, 7] or partial independence in the sense of Weilenmann–Gisin–Sekatski [16]; (ii) defining the witness on a non-product quantity such as a correlation functional involving Alice’s and Charlie’s measurement settings, which survives the Bob-node marginalization; (iii) restricting to specific algebraic structures on the sources, such as J -covariant sources. Each alternative is open. The obstruction is a real no-go for the particular witness L_B , not for the underlying physics.

11 SDP soundness

Theorem 21 (Moment-matrix soundness). *Let $\mathcal{W}$ be a finite set of real polynomial words in operators $\{E_a^x, F_b^y, G_c^z, J_A, J_B, J_C, \dots\}$. For any realization with real density operator ρ , the matrix*

$\Gamma_{u,v} = \text{Tr}(u^\top v \rho)$, $u, v \in \mathcal{W}$, is positive semidefinite.

Proof. For real coefficients $c = (c_u)$, $c^\top \Gamma c = \text{Tr}((\sum_u c_u u)^\top (\sum_v c_v v) \rho) = \text{Tr}(W^\top W \rho) \geq 0$ since $W^\top W \succeq 0$. $\square$

This gives a sound outer-relaxation oracle for leakage-constrained polynomial optimization over states and effects. Convergence of the hierarchy is *not* claimed here. Network-bilocal independence is not enforced by plain PSD moment constraints and requires inflation or scalar-extension constructions (see Renou–Xu–Ligthart [13] and Ligthart–Gachechiladze–Gross [9]); extending those to carry the algebraic constraints $J_X^2 = -I$, $J_X^\top = -J_X$ together with leakage budgets is left to future work.

12 Discussion and limitations

Headline. same- i leakage is a basis-independent compositional monotone whose obstruction to network witnesses is structural rather than technical, and whose geometric quantifier is trace distance rather than positive-noise robustness.

What this paper contributes. A resource theory of compositional complex-structure alignment, with a verified theorem inventory covering: the quotient/sector structure, an affine population monotone, exact free-operations characterizations (dual and Kraus forms), a coherent leakage bound saturated by coherent pure cross-sector states, an explicit trace-distance geometry showing that standard positive-noise robustness diverges, Hamiltonian and Lindblad dynamics formulas, an exactly solved sector-parity discrimination task, the Caves–Fuchs–Rungta worked example, and the Bob-node bilocal obstruction. The companion code package (Section 13) numerically verifies every theorem.

What this paper does not contribute.

1. No falsification of real quantum theory. The resource theory is consistent with both sides of the Renou / Hoffreumon–Woods / Weilenmann–Gisin–Sekatski debate; it quantifies a compositional structure within the real Hilbert-space formalism, independent of whether real QT is operationally distinguishable from complex QT in networks.
2. No proof that complex numbers are necessary. The free face $\mathcal{F}_{AB}$ already recovers the real-vector-space sector of ordinary complex composition.
3. No operational separation from imaginarity in full generality. The sector-parity discrimination task can be saturated by entrywise-real probes. Structurally, L and Hickey–Gour imaginarity differ (Examples 17 and 18), but a single operational task that separates them universally is open.
4. No non-tautological network witness. Proposition 20 fixes the natural Bob-node witness at $1/2$ for every product bilocal source. Operational-independence variants are open.
5. No SDP convergence theorem. Soundness only.

Open problems.

- *Operational separation from imaginarity.* Construct a channel pair $(\mathcal{N}_0, \mathcal{N}_1)$ and probe class where every L -resourceful state outperforms every $\mathcal{I}_B$ -free state, uniformly in basis B .
- *Non-tautological network witness.* Under operational independence (Hoffreumon–Woods), partial independence (Weilenmann–Gisin–Sekatski), or via Alice–Charlie correlation functionals that survive Bob-node marginalization.
- *Convergent leakage-constrained SDP hierarchy.* Extend inflation [9] or scalar-extension [13] hierarchies to carry $J_X^2 = -I$, $J_X^\top = -J_X$, and leakage budgets, and prove convergence.
- *Resource conversion theory.* Asymptotic conversion rates between coherent and population leakage; distillation under free operations.

- *Multiparty alignment.* For N parties, the obvious extension is $P_{\text{phys}} = \prod_{i=2}^N \frac{1}{2}(I - \mathbf{A}_i \mathbf{A}_i)$ giving real dimension $2 \prod_i d_i$. Multiparty monotones, dynamics, and operational tasks are open.

13 Reproducibility

A companion code repository ([GitHub](#)) implements every construction in this paper and runs a pytest suite of 59 tests verifying every theorem numerically. The repository contains: canonical complex structures J_d and the realification map R ; sector operators $\mathbf{A}, \mathbf{B}, M$ and projectors P_+, P_- ; population leakage L , coherent leakage C , alignment weight W , and the trace-distance witness D_1 ; the Kraus-form CPTP monotonicity test $\Phi^\dagger(P_-) \leq P_-$; Hamiltonian and Lindblad simulators verifying conservation conditions and the perturbative $O(\epsilon^2)$ scaling; the sector-parity discrimination optimization with explicit saturating states; the Caves–Fuchs–Rungta closed-form leakage check; the Bob-node obstruction confirmation for product sources; the explicit canonical same- i embedding ι used in Example 18; and the leakage-constrained moment matrix as a CVXPY-ready template. All tolerances are 10^{-10} on algebraic identities and 10^{-3} on the Euler-integrated Lindblad rate. Figures in the manuscript are generated from `figures/make_figures.py`. License: MIT.

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A Theorem inventory

Label	Statement
Theorem 1	Realification preserves statistics, with $\rho_{\mathbb{R}} = \frac{1}{2}R(\rho_{\mathbb{C}})$.
Theorem 2	same- i quotient: $(H_{\mathbb{C}}^A \otimes_{\mathbb{C}} H_{\mathbb{C}}^B)_{\mathbb{R}} \cong K_{AB} / \text{span}\{J_A x \otimes y - x \otimes J_B y\}$.
Theorem 3	Aligned sector $H_+ = \ker(M + I)$ carries the induced complex structure J_{AB} .
Theorem 5	$L \in [0, 1]$, $L = 0 \Leftrightarrow \rho = P_+ \rho P_+$, and L is affine.
Theorem 6	Coherent leakage bound $\ P_+ \rho P_-\ _1 \leq \sqrt{L(1-L)}$.
Theorem 7	Monotonicity iff $\Phi^\dagger(P_-) \leq P_-$.
Theorem 8	Kraus no-leakage iff $P_- K_k P_+ = 0$.
Theorem 9	$D_1(\psi\rangle\langle\psi , \mathcal{F}_{AB}) = \sqrt{\lambda}$ on coherent pure states.
Proposition 10	$W \geq L$; $W = L$ block-diagonally; $W = 1$ on coherent pure.
Proposition 11	Generalized robustness $R = \infty$ when $L > 0$.
Theorem 12	Hamiltonian leakage conservation when $[G, M] = 0$.
Theorem 13	Perturbative leakage $O(\epsilon^2)$.
Theorem 14	Lindblad rate $\dot{L} = \text{Tr}(\mathcal{L}^\dagger(P_-)\rho_t)$.
Corollary 15	$[G, M] = 0 = [R_\ell, M]$ for all ℓ implies exact Lindblad conservation.
Theorem 16	Sector-parity discrimination optimum $\frac{1}{2} + \sqrt{m(1-m)}$.
Proposition 19	CFR leakage $L(\rho_{\text{CFR}}(\alpha)) = (1 - \alpha)/2$.
Proposition 20	Bob-node obstruction $L_B = 1/2$ for product bilocal sources.
Theorem 21	Moment-matrix soundness.